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Problem 1

$\bar{X} = 52 \quad S = 10 \quad n = 100$

- Standard error

$$SE = \frac{S}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1$$

- Test null hypothesis

$H_0: \mu = 50 \quad H_1: \mu \neq 50$

$$z = \frac{\bar{x} - \mu_0}{SE} = \frac{52 - 50}{1} = 2$$

sig level → 10% → $1.645 < 2$ → reject H_0

5% → $1.96 < 2$ → reject H_0

1% → $2.576 > 2$ → fail to reject H_0

- $CI = \bar{X} \pm 1.96(SE) = 52 \pm 1.96(1) = (50.04, 53.96)$

Problem 2

$\hat{S} = 3.1 \quad \hat{SE} = 1.3$ CI = estimate ± critical value × SE

90% → $3.1 \pm 1.645 \times 1.3$

$3.1 \pm 2.1385 = (0.9615, 5.2385)$ reject H_0, $\emptyset$ is not in CI

95% → $3.1 \pm 1.96(1.3) = 3.1 \pm 2.548 = (0.552, 5.648)$ reject H_0, $\emptyset$ is not in CI

99% → $3.1 \pm 2.576(1.3) = 3.1 \pm 3.3488 = (-0.2488, 6.4488)$ fail to reject H_0, $\emptyset$ is in CI

Problem 3

$\bar{X} = 11.5 \quad s = 3 \quad n = 36 \qquad H_0 : \mu = 12 \qquad H_1 : \mu \neq 12$

$$SE = \frac{3}{\sqrt{36}} = \frac{3}{6} = 0.5$$

$$z = \frac{11.5 - 12}{0.5} = -1$$

95% CI $= 11.5 \pm 1.96(0.5) = 11.5 \pm 0.98$

$(10.52, 12.48)$ We fail to reject the null hypothesis.

Problem 4

Group A : $\bar{X}_A = 25, \quad S_A = 8, \quad n_A = 100$

Group B : $\bar{X}_B = 22, \quad S_B = 7, \quad n_B = 100$

Difference in means $= 25 - 22 = 3$

$$SE = \sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}} = \sqrt{\frac{8^2}{100} + \frac{7^2}{100}} = \sqrt{\frac{64}{100} + \frac{49}{100}}$$

$$= \sqrt{1.13} = 1.063$$

$$z = \frac{(\bar{X}_A - \bar{X}_B) - 0}{SE} = \frac{3}{1.063} = 2.82$$

95% CI $= 3 \pm 1.96(1.063)$

$3 \pm 2.083 \rightarrow (0.917, 5.083)$

- We reject the null hypothesis since the CI does not contain zero.

Problem 5

Treatment A: $\bar{X}_A = 14$, $S_A = 6$, $n_A = 81$

Treatment B: $\bar{X}_B = 12$, $S_B = 5$, $n_B = 64$

$\bar{X}_A - \bar{X}_B = 14 - 12 = 2$

$SE = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = \sqrt{\frac{36}{81} + \frac{25}{64}} = \sqrt{0.444 + 0.391} = 0.914$

$Z = \frac{2}{0.914} = 2.19$ $\quad 2.19 > 1.96$, at 5% significance level, we reject the null hypothesis

Problem 6

Factory A: $\bar{X}_A = 3.2$, $S_A = 1.5$, $n_A = 49$

Factory B: $\bar{X}_B = 2.8$, $S_B = 1.4$, $n_B = 64$

$3.2 - 2.8 = 0.4 = \bar{X}_A - \bar{X}_B$

$SE = \sqrt{\frac{1.5^2}{49} + \frac{1.4^2}{64}} = 0.277$

$Z = \frac{0.4}{0.277} = 1.444 < 1.96$ We fail to reject the null hypothesis at a 5% significance level because the CI contains zero and the z-score is less than 1.96.

$CI = 0.4 \pm 1.96(0.277)$

0.4 ± 0.54292

$(-0.14292, 0.94292)$